

Mohamed Mahmoud Samy

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Portfolio & Projects: mohamedmsamy.github.io

SUMMARY

Third-year Mechatronics student at Ain Shams University with hands-on experience in robotic arm control, embedded firmware, and autonomous systems. I've built and deployed distributed embedded systems, simulated robots and controllers in Simscape, and developed autonomous navigation algorithms in ROS2. I like working on systems that have to work in the real world.

EDUCATION

B.Sc. Mechatronics Engineering

Ain Shams University · Expected 2027 · GPA: 3.25

EXPERIENCE

Autonomous Systems Engineer | *F1TENTH Racing Team*

September 2025 – Present · Ain Shams University

- Developed autonomous navigation algorithms for ICRA 2026 AutoDRIVE RoboRacer Sim-Racing competition using LiDAR-based perception in the AutoDRIVE / Isaac Sim environment.
- Implemented and tuned Follow-The-Gap, Pure Pursuit, and Model Predictive Control algorithms for real-time obstacle avoidance and trajectory tracking.
- Collaborated with localization and perception sub-teams to integrate a real-time autonomous driving architecture in ROS2.

FPGA Design Intern | *EgSA – Egyptian Space Agency*

January 2025 – February 2025 · New Administrative Capital, Egypt

- Designed and implemented FPGA modules for embedded aerospace systems including finite state machines, hierarchical components, and arithmetic logic units in VHDL
- Performed RTL simulation, functional verification, and timing analysis to validate designs prior to hardware deployment.

Engineering Intern | *Ghabbour Auto*

July 2024 – August 2024 · Egypt

- Disassembled and inspected engine, manual transmission, and brake systems at the service facility.
- Performed precision measurements using dial gauges, cylinder bore gauges, straight edges, and Vernier calipers including crankshaft runout, engine block flatness, cylinder bore diameter, and brake drum diameter.

CERTIFICATIONS

FPGA & VHDL Workshop - EgSA – Egyptian Space Agency (2025)

MATLAB & Simulink: Onramp + Modeling & Simulation - MathWorks (2025)

Simscape Onramp - MathWorks (2026)

SOLIDWORKS 3D CAD for Education - Dassault Systèmes (2025)

Python for Everybody - University of Michigan (2026)

PCB Design with Eagle - Packt (2025)

SKILLS

Robotics & Simulation: ROS2, MoveIt2, Isaac Sim / AutoDRIVE, Simscape, Gazebo, URDF

Embedded Systems: STM32, ESP32, FreeRTOS, I2C / UART / SPI, FPGA / VHDL

Programming: C / C++, Python, MATLAB / Simulink, VHDL, Java

Hardware & Design: PCB Design, SolidWorks / CAD, Fusion, Circuit Simulation, Stress Analysis

Tools: Git, Linux, RViz, PLC, Eagle